

User Story Mapping

SauceDemo E-commerce Platform

Project: SauceDemo Test Automation

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1. Overview

This document shows how SauceDemo website functions are organized into user stories. These stories help to understand what users do on the website and what needs to be tested. Each story has clear acceptance criteria that will be used to create test cases.

2. User Types

2.1 Different Users

Standard User (`standard_user`)

- Normal customer who can use all website functions
- Can login, browse products, buy items
- Represents most website visitors

Problem User (`problem_user`)

- User who has some technical problems with website
- Some functions might not work correctly
- Helps test error situations

Performance User (`performance_glitch_user`)

- User with slow internet connection or old computer
- Website works slower for this user
- Tests performance problems

Locked User (`locked_out_user`)

- User who cannot access the website
- Login is blocked for security reasons
- Tests access control and security

3. User Journey

Basic path that users follow on the website:

Login → Look at Products → Add to Cart → Check Cart → Pay for Order → Complete Purchase

4. Main Functions and User Stories

Function Group 1: User Login and Access

Feature 1.1: User Login

Goal: Users need to login to access the shopping website.

Story 1.1.1: Successful Login

- **Story:** As a standard user, the system should allow login with correct username and password so that the product page opens.
- **Acceptance Criteria:**
 - User can type username and password
 - System checks if login details are correct
 - User goes to product page after successful login
 - User session starts working
- **Priority:** High

Story 1.1.2: Failed Login - Wrong Details

- **Story:** As a user, the system should show clear error message when wrong username or password is entered so that the problem is understood.
- **Acceptance Criteria:**
 - System shows error message for wrong username/password
 - User stays on login page
 - Error message is easy to understand
 - Login form stays available for retry
- **Priority:** High

Story 1.1.3: Failed Login - Blocked User

- **Story:** As a system, blocked users should not be able to access the website so that security is maintained.
- **Acceptance Criteria:**
 - Blocked user gets specific error message
 - User cannot access the website
 - Error message explains why access is blocked
- **Priority:** High

Story 1.1.4: Failed Login - Empty Fields

- **Story:** As a user, the system should show validation messages when trying to login without entering username or password so that required information is clear.
- **Acceptance Criteria:**
 - System checks that fields are not empty
 - Clear error messages appear
 - Form prevents submission without required data
- **Priority:** Medium

Feature 1.2: Session Management

Goal: User session should work correctly during website use.

Story 1.2.1: Session Keeps Working

- **Story:** As a logged-in user, the session should stay active during shopping so that re-login is not needed.
- **Acceptance Criteria:**
 - Session keeps user logged in
 - Shopping cart contents are saved during session
 - User can move between pages without problems
- **Priority:** Medium

Story 1.2.2: Logout Function

- **Story:** As a user, there should be a way to logout safely so that the session ends properly.
- **Acceptance Criteria:**
 - Logout button is available
 - Session ends when logout is clicked
 - User returns to login page
- **Priority:** Medium

Function Group 2: Product Catalog

Feature 2.1: Product Display

Goal: Users should be able to see products clearly and get information about them.

Story 2.1.1: Product List Display

- **Story:** As a user, all available products should be visible so that shopping choices are clear.
- **Acceptance Criteria:**
 - All products appear in organized layout
 - Product images are visible
 - Product names are shown
 - Product prices are displayed

- Product descriptions are available
- **Priority:** High

Story 2.1.2: Product Details View

- **Story:** As a user, detailed product information should be available so that buying decisions can be made.
- **Acceptance Criteria:**
 - Product image can be viewed larger
 - Complete product description is shown
 - Price is clearly displayed
 - Add to cart button is available
- **Priority:** High

Story 2.1.3: Product Sorting

- **Story:** As a user, products should be sortable by different options so that preferred products are easier to find.
- **Acceptance Criteria:**
 - Sort by name (A-Z, Z-A) works
 - Sort by price (low to high, high to low) works
 - Sorting function works correctly
 - Current sort order is clearly shown
- **Priority:** Medium

Story 2.1.4: Product Information Consistency

- **Story:** As a user, product information should be the same on all pages so that confusion is avoided.
- **Acceptance Criteria:**
 - Product details match on all pages
 - Prices are the same everywhere
 - Images match product descriptions
 - No missing information or broken images
- **Priority:** Medium

Feature 2.2: Product Navigation

Goal: Users should be able to move around the website easily.

Story 2.2.1: Menu Navigation

- **Story:** As a user, the navigation menu should be accessible so that movement between website sections is easy.
- **Acceptance Criteria:**
 - Menu button is easy to find
 - Menu items have clear names

- Navigation links work properly
- Menu works on different screen sizes
- **Priority:** Medium

Story 2.2.2: Back to Products

- **Story:** As a user, there should be a way to return to product list from any page so that shopping can continue.
- **Acceptance Criteria:**
 - Back to products link is available
 - Navigation keeps user context
 - Previous sorting/filtering is remembered
- **Priority:** Medium

Function Group 3: Shopping Cart

Feature 3.1: Add to Cart

Goal: Users should be able to add products to shopping cart for purchase.

Story 3.1.1: Add Single Product

- **Story:** As a user, there should be a way to add a product to shopping cart so that it can be purchased later.
- **Acceptance Criteria:**
 - Add to cart button is available on product list
 - Add to cart button is available on product detail page
 - Cart counter updates immediately
 - User gets confirmation that product was added
- **Priority:** High

Story 3.1.2: Add Multiple Products

- **Story:** As a user, multiple products should be addable to cart so that they can be purchased together.
- **Acceptance Criteria:**
 - Multiple products can be added to cart
 - Cart counter shows total number of items
 - Each product appears separately in cart
- **Priority:** High

Story 3.1.3: Add to Cart Validation

- **Story:** As a user, the cart actions should work reliably so that shopping experience is good.
- **Acceptance Criteria:**
 - Adding same product twice is handled correctly

- Cart state stays consistent when moving between pages
- Add to cart button shows correct state
- **Priority:** Medium

Feature 3.2: Cart Management

Goal: Users should be able to manage their shopping cart contents.

Story 3.2.1: View Cart Contents

- **Story:** As a user, cart contents should be viewable so that purchase items are clear.
- **Acceptance Criteria:**
 - Cart page shows all added products
 - Product details are visible (name, price, description)
 - Quantity information is displayed
 - Total cart value is calculated
- **Priority:** High

Story 3.2.2: Remove Items from Cart

- **Story:** As a user, items should be removable from cart so that unwanted products can be taken out.
- **Acceptance Criteria:**
 - Remove button is available for each item
 - Item disappears immediately when removed
 - Cart counter updates
 - Total price is recalculated
- **Priority:** High

Story 3.2.3: Cart Persistence

- **Story:** As a user, cart contents should stay saved during website use so that selections are not lost.
- **Acceptance Criteria:**
 - Cart contents stay when moving between pages
 - Cart state is kept during user session
 - Cart is accessible from any page
- **Priority:** Medium

Story 3.2.4: Empty Cart Handling

- **Story:** As a user, empty cart state should be clear so that next actions are understood.
- **Acceptance Criteria:**
 - Empty cart message is displayed
 - Continue shopping option is available
 - Cart counter shows zero
- **Priority:** Low

Feature 3.3: Cart Navigation

Goal: Users should be able to access cart easily from anywhere on website.

Story 3.3.1: Cart Access

- **Story:** As a user, shopping cart should be accessible from any page so that cart contents can be checked.
- **Acceptance Criteria:**
 - Cart icon/link is available on all pages
 - Cart badge shows number of items
 - Cart can be opened with one click
- **Priority:** High

Story 3.3.2: Continue Shopping

- **Story:** As a user, there should be a way to continue shopping from cart so that more items can be added.
- **Acceptance Criteria:**
 - Continue shopping button is available
 - Button returns to product catalog
 - Cart contents are saved
- **Priority:** Medium

Function Group 4: Checkout Process

Feature 4.1: Checkout Information

Goal: Users should be able to provide information needed for purchase.

Story 4.1.1: Checkout Information Form

- **Story:** As a user, there should be a form to enter checkout information so that purchase can be completed.
- **Acceptance Criteria:**
 - First name field is available and required
 - Last name field is available and required
 - Postal code field is available and required
 - Form validation prevents submission with missing fields
 - Clear error messages appear for validation problems
- **Priority:** High

Story 4.1.2: Checkout Information Validation

- **Story:** As a user, checkout information should be validated so that errors are prevented.
- **Acceptance Criteria:**
 - Required field validation works
 - Format validation works where needed
 - Clear error messages are shown
 - Form can only be submitted with valid data
- **Priority:** High

Story 4.1.3: Checkout Navigation

- **Story:** As a user, navigation through checkout process should be clear so that purchase can be completed easily.
- **Acceptance Criteria:**
 - Cancel button returns to cart
 - Continue button moves to order review
 - Back navigation is available
 - Form data is saved during navigation
- **Priority:** Medium

Feature 4.2: Order Review

Goal: Users should be able to review order before completing purchase.

Story 4.2.1: Order Overview Display

- **Story:** As a user, order summary should be visible so that purchase can be verified before payment.
- **Acceptance Criteria:**
 - All cart items are displayed
 - Individual item prices are shown
 - Subtotal is calculated correctly
 - Tax is calculated and displayed
 - Total price is calculated correctly
- **Priority:** High

Story 4.2.2: Order Modification from Overview

- **Story:** As a user, order should be changeable from overview page so that last-minute changes can be made.
- **Acceptance Criteria:**
 - Cancel button returns to cart
 - Back button returns to information form
 - User can modify quantities or remove items
- **Priority:** Medium

Story 4.2.3: Price Calculation Accuracy

- **Story:** As a user, price calculations should be accurate so that payment amount is trusted.
- **Acceptance Criteria:**
 - Item prices match catalog prices
 - Subtotal equals sum of item prices
 - Tax calculation is correct
 - Total is sum of subtotal and tax
- **Priority:** High

Feature 4.3: Order Completion

Goal: Users should be able to complete purchase successfully.

Story 4.3.1: Order Completion

- **Story:** As a user, there should be a way to complete order so that purchase is finalized.
- **Acceptance Criteria:**
 - Finish button completes the order
 - Order confirmation page is displayed
 - Success message is shown
 - Order details are summarized
- **Priority:** High

Story 4.3.2: Order Confirmation

- **Story:** As a user, order confirmation should be provided so that purchase success is clear.
- **Acceptance Criteria:**
 - Confirmation page displays order details
 - Thank you message is shown
 - Order reference is provided
 - Next steps are explained
- **Priority:** High

Story 4.3.3: Post-Purchase Navigation

- **Story:** As a user, navigation after purchase should be available so that website use can continue.
- **Acceptance Criteria:**
 - Back to products button is available
 - Cart is cleared after successful purchase
 - User can start new shopping session
- **Priority:** Medium

Function Group 5: User Experience

Feature 5.1: Responsive Design

Goal: Website should work well on different devices and browsers.

Story 5.1.1: Mobile Responsiveness

- **Story:** As a mobile user, the website should work well on mobile device so that shopping is possible from mobile.
- **Acceptance Criteria:**
 - Layout adapts to mobile screen sizes
 - Touch interactions work properly
 - Text is readable without zooming
 - Navigation is accessible on mobile
- **Priority:** Medium

Story 5.1.2: Cross-Browser Compatibility

- **Story:** As a user, the website should work consistently in different browsers so that preferred browser can be used.
- **Acceptance Criteria:**
 - Functions work in Chrome, Firefox, Edge
 - Visual appearance is consistent across browsers
 - No browser-specific errors occur
- **Priority:** Medium

Feature 5.2: Error Handling

Goal: Users should get clear information when something goes wrong.

Story 5.2.1: User-Friendly Error Messages

- **Story:** As a user, clear error messages should appear when something goes wrong so that the problem is understood.
- **Acceptance Criteria:**
 - Error messages are clear and helpful
 - Technical language is avoided
 - Next steps are provided where possible
- **Priority:** Medium

Story 5.2.2: Error Recovery

- **Story:** As a user, recovery from errors should be easy so that website use can continue.
- **Acceptance Criteria:**
 - Clear path to fix errors is provided
 - Option to retry failed actions is available
 - Session state is kept when possible

- **Priority:** Medium
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Function Group 6: Security and Performance

Feature 6.1: Security Measures

Goal: Website should be secure and protect user information.

Story 6.1.1: Input Validation

- **Story:** As a user, input should be validated so that website security is maintained.
- **Acceptance Criteria:**
 - All form inputs are validated
 - Dangerous input is prevented
 - Security measures work properly
- **Priority:** High

Story 6.1.2: Session Security

- **Story:** As a user, session should be secure so that account is protected.
- **Acceptance Criteria:**
 - Session tokens are managed properly
 - Session timeout works correctly
 - Secure logout function is available
- **Priority:** Medium

Feature 6.2: Performance Requirements

Goal: Website should work fast and smoothly.

Story 6.2.1: Page Load Performance

- **Story:** As a user, pages should load quickly so that website use is efficient.
- **Acceptance Criteria:**
 - Page load time is under 3 seconds
 - Images load efficiently
 - No significant performance problems
- **Priority:** Medium

Story 6.2.2: Performance Under Load

- **Story:** As a performance-sensitive user, website should work well even with slow connection.
- **Acceptance Criteria:**
 - Website stays functional during performance issues
 - User feedback is provided during slow operations
 - Performance degrades gracefully
- **Priority:** Low

5. Summary

5.1 Test Coverage Plan

Each user story will create multiple test scenarios:

- **Positive scenarios:** Normal user behavior testing
- **Negative scenarios:** Error handling and edge cases testing
- **Boundary scenarios:** Limit and constraint testing
- **Integration scenarios:** Complete workflow testing

Expected Test Cases: 50+ (at least 1-2 test cases per user story, complex stories will have 3-5 test cases)

5.2 Requirements to Test Cases Connection

Each acceptance criterion will be connected directly to specific test cases. This ensures:

- 100% requirement coverage
 - Clear connection from requirements to tests
 - Complete validation of all functions
 - Both positive and negative test scenarios
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6. Next Steps

1.  **Next: Test Case Development** - Based on this story mapping
2. **Pending: Test Framework Design**
3. **Pending: Framework Implementation**

This user story mapping provides the foundation for creating detailed test cases that will ensure complete coverage of SauceDemo functions while staying aligned with business requirements and user expectations.
